Reusable Social Trust Predictor

EVS/WVS Joint 2017-2022 | Python prediction tool | Production deliverable

Purpose and audience

A reusable Python tool for students and researchers exploring whether a survey respondent would answer **'most people can be trusted'**. It applies the tuned multilayer perceptron (MLP) classifier from Part II to a **new respondent**. It predicts a category and reports the model's estimated positive-class probability; it does not estimate the causal effects of personal characteristics.

Inputs, outputs and implementation

The input is a one-row pandas DataFrame using the original 12 survey-coded variables: age (**X003**), sex, education, employment, marital status, health, sports and charity membership, religious status, freedom of choice, importance of friends, and country (**cntry**). The output is **0** ('cannot be too careful') or **1** ('most people can be trusted'), with a score for class 1.

The saved **scikit-learn Pipeline** includes training-fitted missing-value imputation, category encoding, feature scaling and the MLP. The separate **social_trust_predictor.py** wrapper checks required inputs, survey coding ranges and supported countries; it rejects the target variable **A165** as an input. Missing age can be imputed using the training-fitted median. The country list is stored in **supported_countries.json**.

Using the tool

Keep the **.py**, **.joblib**, country JSON, README and requirements files together. Install the pinned versions with **pip install -r requirements.txt**. The README explains all 12 input scales and provides a complete example of a synthetic respondent.

```
from social_trust_predictor import predict_trust

prediction, trust_probability = predict_trust(new_person)
```

Production tests and reproducibility

The accompanying **finalize_part3.py** runs independently of the training notebook and writes **production_test_results.txt**. Its nine checks cover: equality of wrapper and saved-model class/score; repeatability; missing-age imputation; refusal of out-of-range age and education; refusal of unknown country and negative survey codes; rejection of a missing required column; and prevention of target leakage. **Packaging is blocked if any check fails.** The student's earlier notebook checks also compared saved and original fitted-model predictions.

The final ZIP contains the trained pipeline, source code, country codes, README, pinned dependency versions, test evidence and this PDF. It requires no raw survey records, external account or retraining step. Serialized joblib models should only be loaded from trusted sources.

Limits and responsible use

The probability is a model estimate, **not a demonstrated calibrated probability**. Self-reported survey variables, selection effects and differences across countries limit transfer to new populations. This predictor does not establish causal effects and should not be used for high-stakes individual decisions. If deployed, monitor country-level errors, new input patterns and calibration; retrain when justified by newer appropriately licensed data.

Source: European Values Study / World Values Survey, Joint EVS/WVS 2017-2022, version 5.0 (ZA7505). Production acceptance evidence: production_test_results.txt generated on the user's machine.